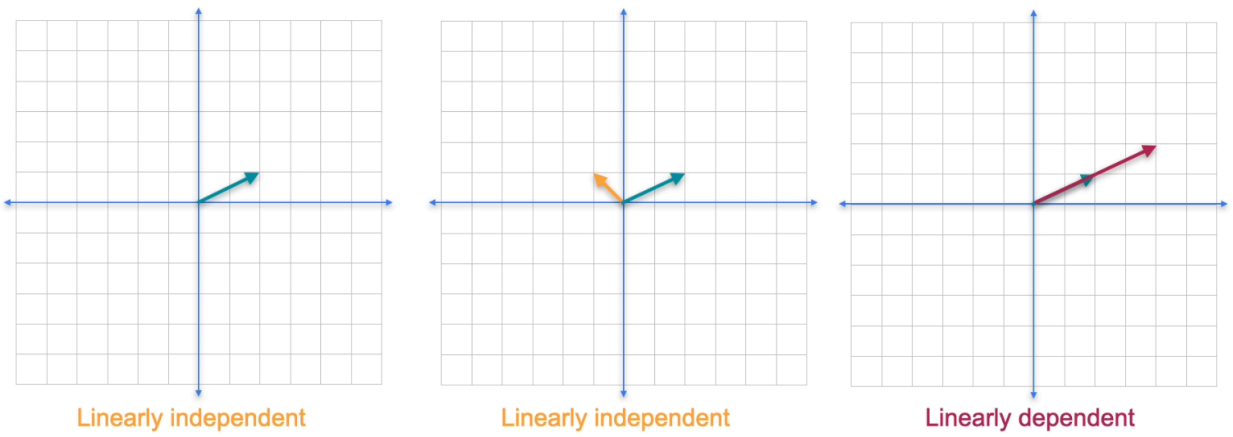
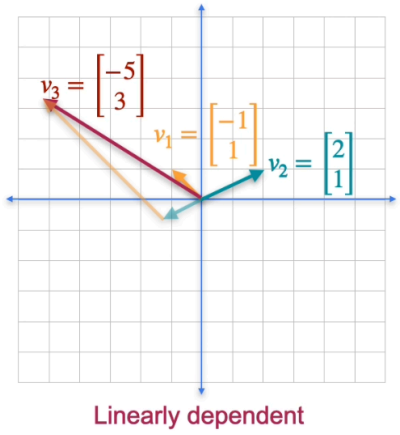
Linear Dependence and Independence

Linear Dependence = the matrix is singular if and only if its rows or columns are linearly dependent.



Check for linear dependence



$$\alpha v_1 + \beta v_2 = v_3$$

$$\alpha \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$$

1 $\quad -\alpha + 2\beta = -5$

2 $\quad \alpha + \beta = 3$

1 + 2

$$3\beta = -2 \longrightarrow \beta = -\frac{2}{3}$$

2

$$\alpha - \frac{2}{3} = 3 \longrightarrow \alpha = \frac{11}{3}$$